

Weighted V3 Attitude Scoring: Mathematical Breakdown

This document provides the exact mathematical calculations used to determine the final score for every single participant response in the “Weighted V3” attitude analysis.

The algorithm uses an inverse-frequency probability calculation to scale the rewards and penalties.

Mathematical Formula:

Final Weighted Score = Assigned Weight * (1 - p) (Where *p* is the proportion of the 201 participants who gave that exact answer).

Q28. What should a thalassemia carrier do after diagnosis? (Tick all that apply)

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
Get The Partner Tested Before Marriage	2	0.791	0.209	0.418
Get Family Members Tested	1	0.786	0.214	0.214
I Don't Know	0	0.075	0.925	0.000
Ignore It	-3	0.000	1.000	-3.000

Q30. Are you willing to / Do you have a consanguineous marriage?

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
Definitely Not	2	0.796	0.204	0.408
Not Sure	-1	0.174	0.826	-0.826
Yes I Am Willing	-3	0.015	0.985	-2.955
Yes I Have	-3	0.010	0.990	-2.970

Q31. Do you accept marriage between two thalassemia carriers?

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
No	2	0.935	0.065	0.129
Not Sure	-1	0.164	0.836	-0.836
Yes	-3	0.050	0.950	-2.851

Q32. How important is thalassemia screening before marriage?

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
Very Important	2	0.672	0.328	0.657
Important	1	0.925	0.075	0.075
Not Sure	-1	0.065	0.935	-0.935
Not Important	-2	0.020	0.980	-1.960

Q35. Do you think is it important for your family members to undergo screening?

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
Agree	2	0.970	0.030	0.060
Don't Know	0	0.025	0.975	0.000
Disagree	-2	0.020	0.980	-1.960

Q39. How easy is it to convince relatives to undergo screening?

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
Very Easy	2	0.144	0.856	1.711
Somewhat Easy	1	0.463	0.537	0.537
Not Sure	0	0.164	0.836	0.000
Difficult	-1	0.189	0.811	-0.811

Q40. How important is cascade screening in thalassemia prevention?

Answer Option	Assigned Weight (V3)	Proportion (p)	Multiplier (1-p)	Final Weighted Score
Very Important	2	0.587	0.413	0.826
Important	1	0.985	0.015	0.015
Slightly Important	-1	0.040	0.960	-0.960
Not Important	-2	0.010	0.990	-1.980